

Sigma-Point Kalman Filtering for Drone State Estimation

Description

This project implements a Sigma-Point Kalman Filter (SPKF), specifically a Square-Root Unscented Kalman Filter (UKF), for estimating the lateral motion of a drone. The system dynamics are nonlinear and include coupled position, speed, roll, and heading states. Noisy GPS position measurements are used to recursively estimate the full state over time.

System Model

The state vector is:

$$\underline{x} = \begin{bmatrix} p_x \\ p_y \\ s \\ \phi \\ \psi \end{bmatrix}$$

where (p_x, p_y) is the position of the drone, s is the speed, ϕ is the roll angle, and ψ is the heading angle.

The drone is modeled using coordinated-turn dynamics:

$$\dot{\underline{x}} = \begin{bmatrix} s * \cos\psi & s * \sin\psi & a & \omega & \frac{g}{s} \sin\phi \end{bmatrix}^T + \underline{\nu}$$
$$\underline{\nu} \sim N(0, \mathbf{Q})$$

$$\mathbf{Q} = \text{diag} \left[0.1^2 \quad 0.1^2 \quad 0.1^2 \quad (2\frac{\pi}{180})^2 \quad (2\frac{\pi}{180})^2 \right]$$

where a is the acceleration input, ω is the roll-rate input, and g is acceleration of gravity ($9.8m/s^2$).

The drone has noisy GPS measurements of (p_x, p_y) position:

$$\mathbf{z} = \mathbf{H}\underline{x} + \underline{\eta}$$
$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$
$$\underline{\eta} \sim N(0, \mathbf{R})$$
$$\mathbf{R} = \text{diag} \left[3^2 \quad 3^2 \right]$$

Sigma-Point Kalman Filtering

Sigma-Point Kalmaning Filters approximate the propagation of a Gaussian distribution through non-

linear dynamics using a deterministic set of sample points. These sigma points are chosen to capture the mean and covariance of the state distribution.

Each sigma point is propagated through the nonlinear system, and the transformed points are recombined to recover the mean and covariance. This approach avoids linearization and provides improved accuracy compared to the Extended Kalman Filter.

Unscented Kalman Filter

In this project, the sigma-point framework is implemented using the Unscented transform. The UKF selects sigma points based on the current state estimate and covariance, propagates them through the nonlinear dynamics, and computes weighted estimates of the predicted mean and covariance.

The accuracy of the UKF depends on selecting the optimal values for the parameters α , β , and κ . α controls the spread of sigma points around the mean, β encodes prior knowledge about the distribution, and κ is a secondary scaling parameter.

Square-Root Formulation

A square-root form of the UKF is used in this implementation. Instead of propagating the covariance matrix directly, the filter maintains and propagates its Cholesky factor.

This approach provides several advantages:

- improved numerical stability
- preservation of positive definiteness
- reduced sensitivity to roundoff errors

These properties are especially important in nonlinear estimation problems, where covariance matrices can degrade over time when propagated directly.

Simulation Setup

The filter is run with a sampling period of 0.1 s over 1000 samples. The initial state estimate is

$$\hat{\mathbf{x}} = [0 \ 0 \ 25 \ 0 \ 0]^\top$$

with initial covariance

$$\hat{\mathbf{P}} = \text{diag}([10^2 \ 10^2 \ 3^2 \ (\frac{1}{18})^2 \ (\frac{1}{18})^2])$$

Results

Figure 1 shows the true and estimated drone trajectory.

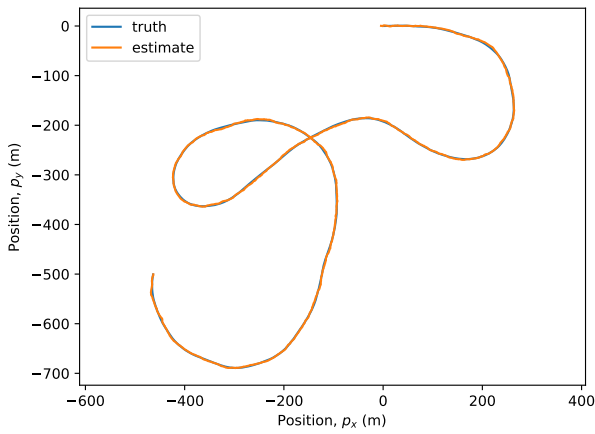


Figure 1: True and estimated drone trajectory

Figures 2–4 show the position, speed, roll, and heading errors over time along with their corresponding $3 - \sigma$ bounds.

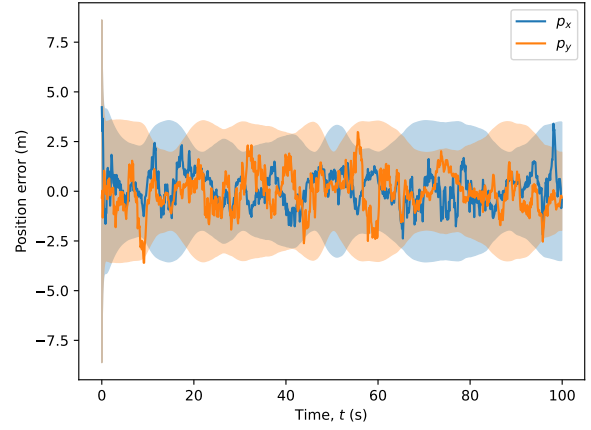


Figure 2: Position errors

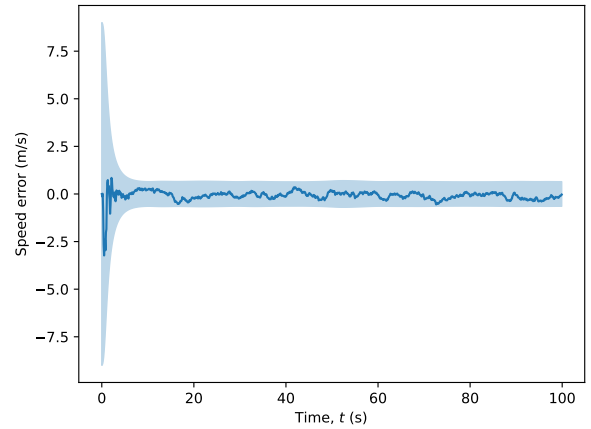


Figure 3: Speed errors

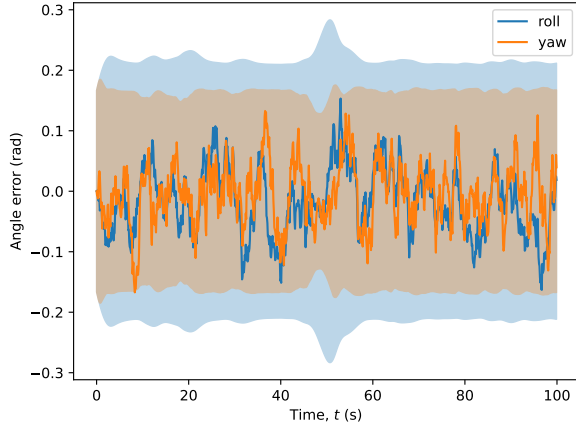


Figure 4: Roll and Heading errors

Discussion

The results demonstrate the sigma-point filtering is effective for nonlinear flight dynamics. Even with limited measurements, the filter is able to infer the full state through the structure of the system dynamics.

The square-root formulation improves numerical robustness and helps maintain a consistent covariance representation over time. This is particularly important for nonlinear systems and longer simulations.

This project highlights the advantage of sigma-point methods: accurate nonlinear estimation without requiring analytical linearization, combined with strong numerical properties when implemented in square-root form.